

Haotong Luan

Shenzhen, China

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Summary

Third-year automation undergraduate working on underactuated robotic hands, environment-adaptive grasping, and mechanism design. Experience spans gripper development, visual-servoing control, and hardware prototyping, with interests in VLMs, VLAs, and WAMs for reliable robotic manipulation.

Education

Shenzhen Technology University – Future Technology School

B.ENG. IN AUTOMATION -- THIRD-YEAR UNDERGRADUATE

Shenzhen, China

Sep. 2024 – Present

- GPA: 84.17/100. Research focus: underactuated grippers, environment-adaptive robotic hands, and mechanism design.

Tsinghua University | Shenzhen X-Institute

P-SRT PROGRAM -- JOINT TRAINING IN UNDERACTUATED ROBOTICS

Shenzhen, China

Sep. 2025 – Present

- Advisor: Associate Prof. Wenzeng Zhang. Current stage: P-SRT; program pathway: X-idea → P-SRT → E-SRT → ORIC → SURF.

Research Experience

Shanghai AI Laboratory

RESEARCH ASSISTANT -- VISUAL-SERVOING CONTROL AND END-EFFECTOR DESIGN

Remote

Apr. 2026 – Present

- Working with Researcher Yuan Gao on visual-servoing control for robotic-arm grasping; responsible for the mechanical design of the arm-mounted gripper.

Shenzhen X-Institute, Tsinghua University

P-SRT RESEARCHER -- UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

Sep. 2025 – Present

- Independently proposed the dual-slide adaptive gripper concept and conducted mechanism analysis and feasibility studies under Associate Prof. Wenzeng Zhang.
- Designed linear-motion linkages, parallelogram mechanisms, and idle-stroke transmissions for reliable single-actuator grasping. Led DS Hand (first author, RAAI 2025); collaborated on idle-stroke and GLINT grippers.

Shenzhen Loop Area Institute

QUANTUM & AI WINTER SCHOOL

Shenzhen, China

Jan. 2026

- Studied MLP/LSTM/Transformer-based quantum variational circuit simulation with Prof. Peng Zhang (Tianjin University), including state fidelity optimization, mixed-state simulation, and physics-constrained training.

Publications

* Corresponding author. Submitted manuscripts are listed separately.

PEER-REVIEWED PUBLICATIONS AND ACCEPTED PAPERS

[P1] DPS Hand: A Dual-Parallelogram and Slot-Guided Gripper for Adaptive Grasping and Environmental Scooping

Haotong Luan, Haokai Ding, Tao Yang*, Wenzeng Zhang*

IEEE CASE 2026

Accepted; RAS major conference; Shenyang, China, Aug. 17–21, 2026.

[P2] PlaceRecover: Layout-Invariant Point Cloud Recovery for Robust 3D Perception under LiDAR Placement Shifts

Benwu Wang, Zihang Wang*, Wenkai Zhu, Haotong Luan, Dong Kong, Binghao Wang, Yiming Zhou, Kailin Lyu, Teng

IEEE/RSJ IROS 2026

Yan, Haoyu Wang, Yaohua Liu, Qiang Zhang

Accepted; top robotics conference; Pittsburgh, PA, USA, Sep. 27–Oct. 1, 2026.

[P3] CSD: Conformal Selective Diagnosis for Reliable Bearing Fault Identification Under Domain Shift

Xi Yu, Anran Lu, Keyi Chen, Haotong Luan*, Yuan Gao*

IEEE SMC 2026

Accepted; CCF C; Bellevue, WA, USA, Oct. 4–7, 2026. Corresponding author.

[P4] Domain-Semantic Subspace Stacking for Financial Fraud Detection

Haotong Luan, Minmin Chen, Zhenhui Lin, Haokai Ding, Yiwei Sheng*

ICIC 2026

Accepted; CCF C; Toronto, Canada, Jul. 22–26, 2026.

[P5] An Underactuated Dual-Slide Adaptive Robotic Hand with Strong Environmental Adaptability

Haotong Luan, Wenzeng Zhang*, Tao Yang*

RAAI 2025

Singapore, Dec. 18–20, 2025. DOI: [10.1109/RAAI67517.2025.11423404](https://doi.org/10.1109/RAAI67517.2025.11423404).

[P6] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Yizhe Chen, Haotong Luan, Haokai Ding, Wenzeng Zhang*

RAAI 2025

Singapore, Dec. 18–20, 2025. DOI: [10.1109/RAAI67517.2025.11423397](https://doi.org/10.1109/RAAI67517.2025.11423397).

[P7] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

Haokai Ding, **Haotong Luan**, Tao Yang*, Wenzeng Zhang*
Paris, France, Nov. 24–26, 2025. DOI: 10.1109/ICCMA67641.2025.11369602.

ICCMA 2025

JOURNAL MANUSCRIPTS SUBMITTED / UNDER REVIEW

[M1] Manuscript submitted to Advanced Intelligent Systems

Haotong Luan, Jie Zhang, Yuan Gao
Intelligent systems and robotics; JCR Q1.

Submitted, 2026

[M2] Manuscript under review at Scientific Reports

Haotong Luan, Xi Yu, Anran Lu, Keyi Chen, Jianwu Chen
JCR Q1.

Under review, 2026

Patents Pending

3 invention patents and 1 utility model patent filed and under review.

Honors & Awards

- 2026 **AIR Summer School Admission**, Institute for AI Industry Research, Tsinghua University.
- 2026 **CASE 2026 IEEE RAS Travel Support** (Travel Grant).
- 2026 **National-Level Project, Principal Investigator**, College Student Innovation and Entrepreneurship Training Program: “Zero-Lash Smart Grasping”.
- 2026 **Provincial-Level Project, Member**, College Student Innovation Training Program: “Multi-Core Fiber Dynamic Optical Trap for Single-Cell Pathology”.
- 2026 **SDG Global Summer School Selection** (tuition waived), Zhejiang University & Harbin Institute of Technology.
- 2026 **Quantum & AI Winter School Selection**, youngest of only 9 undergraduates among 70+ participants.
- 2024–2025 **Dean’s Special Award for Rising Talent, Zhenxiong**, Future Technology School, SZTU.
- 2025 **Crystal Prize for Scientific & Technological Innovation**, top 10 across SZTU.
- 2024, 2025 **Outstanding Student Cadre**, Shenzhen Technology University.
- 2025 **Excellence Prize**, “Challenge Cup” Student Entrepreneurship Competition, university round.
- 2025 **Excellence Prize**, “Challenge Cup” Extracurricular Academic Science and Technology Competition, university round.
- 2025 **Outstanding Volunteer**, 15th National Games of China and 12th National Games for Persons with Disabilities / 9th National Special Olympic Games.

Professional Service & Leadership

- 2026 **Conference Reviewer**, IEEE-RAS International Conference on Humanoid Robots (Humanoids), a top conference in humanoid robotics.
- 2026 **Conference Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), a top robotics conference.
- 2026 **Workshop Reviewer**, IAB @ NeurIPS 2026 Workshop.
- 2024–Present **Head of Organization Department**, Future Technology School, SZTU (since Dec. 2024).
- 2024–2025 **Class Monitor & Youth League Secretary**, Sep. 2024–Oct. 2025.

Skills

Mechanism Design	Parallel-jaw coupled grippers, linear linkages, parallelogram linkages, underactuated and tendon-driven hands; SolidWorks (modeling, assembly, kinematic simulation); Bambu Lab 3D printing; prototyping and grasping experiments.
Programming	Python; basic deep learning for point clouds and quantum circuit simulation.
Academic Writing	IEEE journal/conference formatting and submission; AI-assisted literature review.
Languages	Chinese (native); English (CET-4: 580; IELTS scheduled Dec. 2026).